

**SULIT**



## **SECP2523 DATABASE**

**SESSION 2024/2025 - SEMESTER 1**

### **ALTERNATIVE ASSESSMENT REPORT: PHASE 1**

|                            |                             |
|----------------------------|-----------------------------|
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| <b>Case Study</b>          | WBL PROJECT                 |
| <b>System Name</b>         | KOPERASI KADA SYSTEM        |
| <b>Group Name</b>          | TECHNICRAB                  |

|                                                 |                                          |                                                                                                                                                                              |
|-------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SECTION A (SYSTEM'S OVERVIEW)</b>            |                                          |                                                                                                                                                                              |
| 1.1                                             | <b>Overall Description</b>               |                                                                                                                                                                              |
|                                                 |                                          | << <i>Describe the system's overview.</i> >>                                                                                                                                 |
|                                                 |                                          |                                                                                                                                                                              |
| 1.2                                             | <b>Project Weaknesses or Improvement</b> |                                                                                                                                                                              |
|                                                 |                                          | << <i>List the project's weaknesses or improvement that can be made to the proposed system. Provide an explanation of the proposed module to overcome the weaknesses.</i> >> |
| 1.3                                             | <b>Proposed Module</b>                   |                                                                                                                                                                              |
|                                                 |                                          | << <i>Describe a new module to overcome the project's weaknesses.</i> >>                                                                                                     |
|                                                 |                                          |                                                                                                                                                                              |
| <b>SECTION B (DATABASE PLANNING AND DESIGN)</b> |                                          |                                                                                                                                                                              |
| 2.1                                             | <b>Conceptual ERD</b>                    |                                                                                                                                                                              |
|                                                 |                                          | << <i>Draw a conceptual ERD to represent the proposed module.</i> >>                                                                                                         |
|                                                 |                                          |                                                                                                                                                                              |
| 2.2                                             | <b>Logical ERD</b>                       |                                                                                                                                                                              |
|                                                 |                                          | << <i>Provide logical ERD of your module in the proposed system.</i> >>                                                                                                      |
|                                                 |                                          |                                                                                                                                                                              |
| 2.3                                             | <b>Data Dictionary for Logical ERD</b>   |                                                                                                                                                                              |
|                                                 |                                          | << <i>Provide the data dictionary based on the logical ERD in Subsection 2.2.</i> >>                                                                                         |
|                                                 |                                          |                                                                                                                                                                              |
| 2.4                                             | <b>Relational Database Schema</b>        |                                                                                                                                                                              |
|                                                 |                                          | << <i>Provide all relations database schema of your proposed module.</i> >>                                                                                                  |
|                                                 |                                          |                                                                                                                                                                              |
| 2.5                                             | <b>Normalization</b>                     |                                                                                                                                                                              |
|                                                 |                                          | << <i>Illustrate the normalization process up until BCNF to every relation listed in 2.5. At the end of this section, list all normalized relations achieved.</i> >>         |

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## Section A: System Overview

### 1.1 Overview Description

The Koperasi KADA system is to create a fully integrated digital platform that enhances the efficiency, security and user experience for all members and administrators. The system is a web-based platform that designed to digitize and optimize the internal processes of Koperasi KADA. It focuses on member registration, loan applications and financial management. This system provides an integrated interface for members and administrators to enable seamless interaction and process automation.

The system enables prospective employees to register online, submit required documents and track their application status in real time. The approved members can access additional features such as applying for loans, managing savings and viewing financial statements. Meanwhile, the system centralizes member and loan data for administrators by allowing them to process with applications, approve or reject member registration and loan requests. In the system, the administrators also can generate comprehensive reports efficiently. The automated workflows and secure data handling can minimize human error and reduce dependency on manual paperwork.

The Koperasi KADA system is a user-friendly interface and role-based access controls ensure that different roles of user can navigate the system effortlessly while accessing only the functionalities pertinent to their role. Overall, the system serves as a transformative tool, fostering increased productivity, improved service delivery and enhanced member satisfaction through its efficient, secure and integrated platform.

| Group Member's Name    | Matric No | Module                                                                    |
|------------------------|-----------|---------------------------------------------------------------------------|
| Lau Yee Wen            | A23CS0099 | Employee Module (apply membership and view membership application status) |
| Chau Ying Jia          | A23CS0213 | Member Module (apply for loan and view loan application status)           |
| Cheryl Cheong Kah Voon | A23CS0060 | Member Module (view financial status and manage profile)                  |
| Neo Li Xin             | A23CS0253 | Admin Module (generate reports)                                           |

### 1.2 Project Weakness or Improvement

The system currently lacks a robust automated alert mechanism to notify administrators about pending registration and loan application approvals. This deficiency increases the risk of important tasks being overlooked or delayed, particularly when administrators manage a high volume of applications. Without timely alerts, administrators may miss critical deadlines, leading to prolonged waiting periods for applicants. This delay can cause frustration among applicants, as they are left uncertain about the status of their requests.

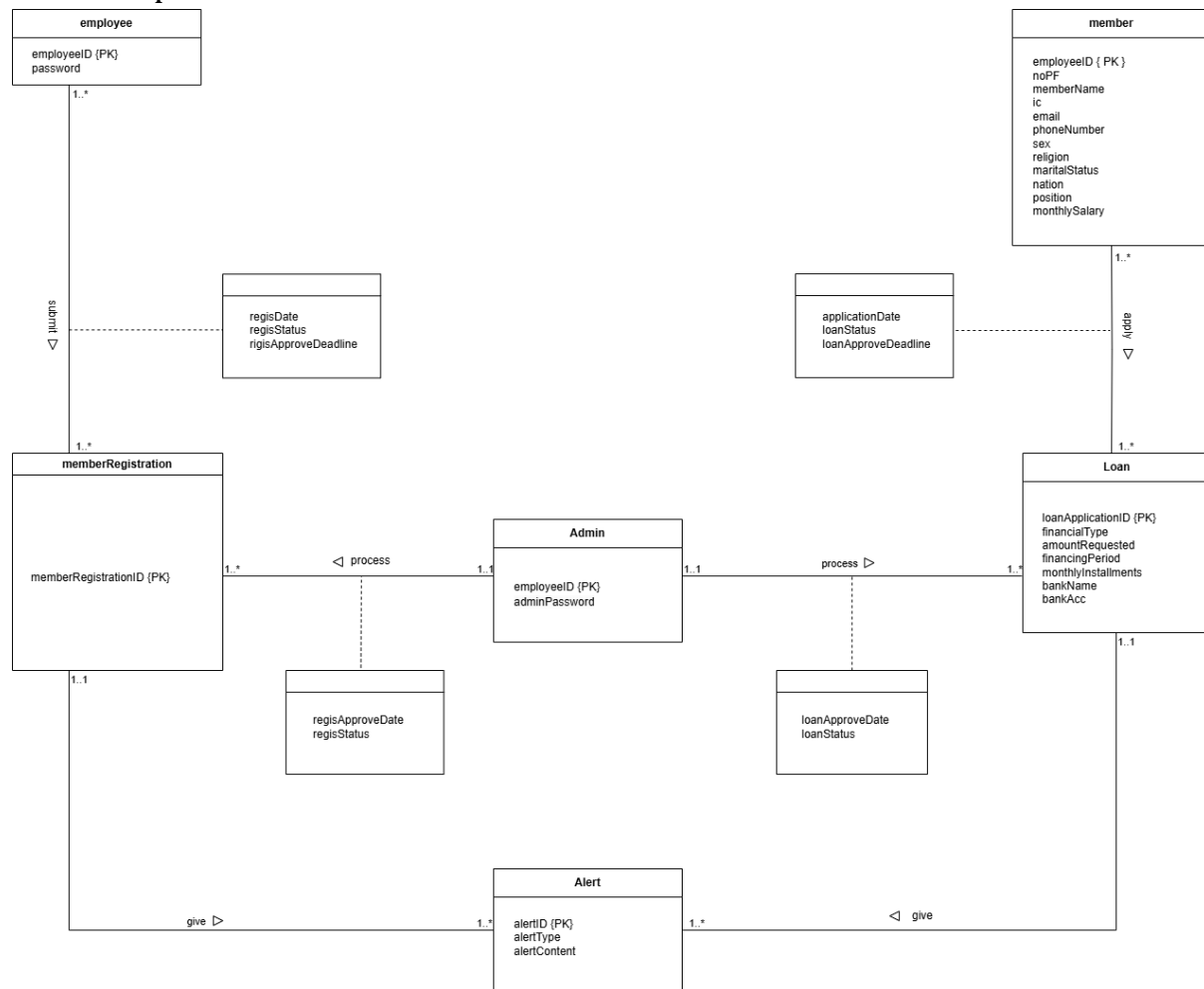
### 1.3 Proposed Module

Based on the answer in Section A(2), the new module to overcome the weaknesses is an automated notification and alert system. This module aims to introduce automated notifications and alerts for both administrators, users and members. It ensures timely updates and efficient communication between all stakeholders involved in the registration and loan application process.

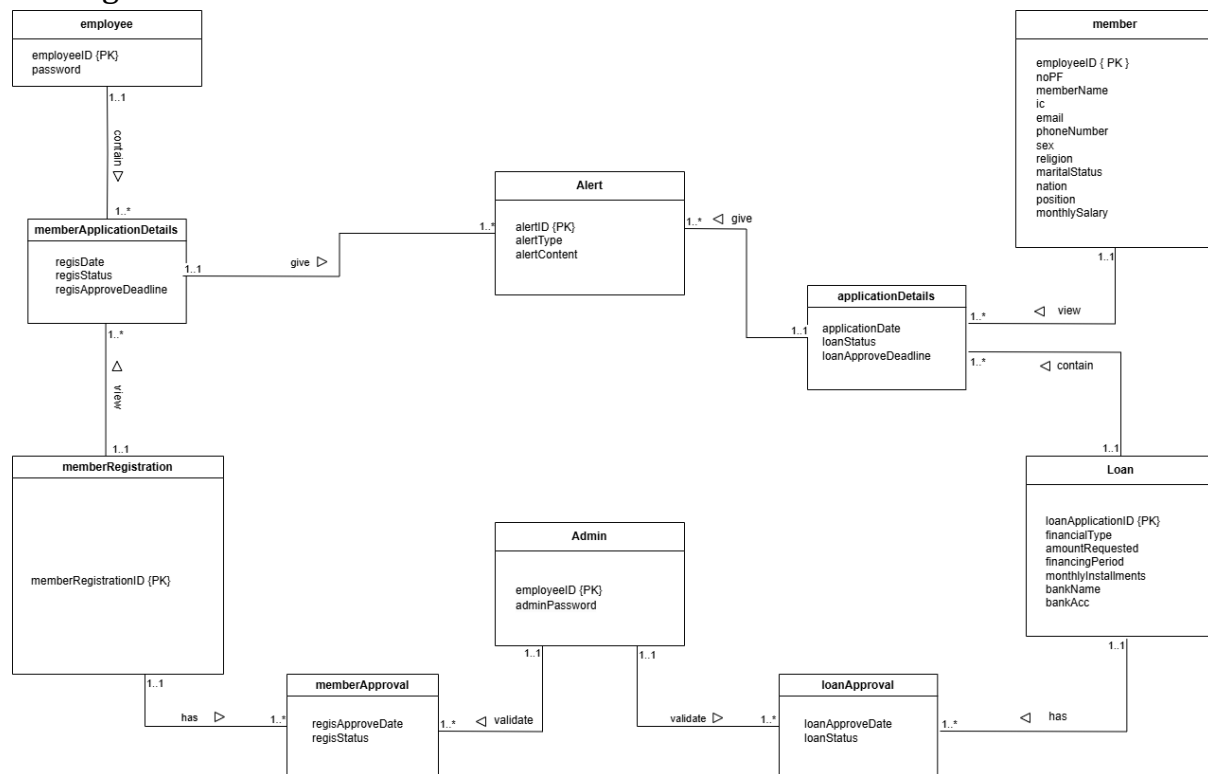
The system will send reminders to administrators about pending approvals for registrations and loan applications automatically after a setting period. Meanwhile, the system will notify users or members immediately when their registration or loan application status changes or requires additional information. The notification can provide detailed instructions in notifications if further actions are needed. The notification and alert can be sent via email, SMS or an in-system notification panel.

## Section B: Database Planning and Design

## 2.1 Conceptual ERD



## 2.2 Logical ERD



## 2.3 Data Dictionary for Logical ERD

### 2.3.1 Identifying Entity

| Entity Name               | Description                                                          | Aliases              | Occurrence                                                        |
|---------------------------|----------------------------------------------------------------------|----------------------|-------------------------------------------------------------------|
| Employee                  | Represents any person accessing the KADA system                      | New user, guest      | Occurs when first-time user visit the website                     |
| Member                    | Represents registered users who have created KADA accounts           | Members              | Occurs when new user or guest register as a member                |
| Admin                     | Represents the system admin who responsible for managing KADA system | Administrator, staff | Occurs when system built and processing members' account activity |
| memberRegistration        | Describe all activities when a user register as a member             | Users                | Occurs when a new user register as a member through KADA system   |
| memberApplication Details | Contains detailed information about the                              | Registration details | Occurs when a member fills out detailed information               |

|                    |                                                                   |                          |                                                                               |
|--------------------|-------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------|
|                    | member's registration application                                 |                          | during the registration process                                               |
| memberApproval     | Represents the approval process for member registrations          | Registration approval    | Occurs when the admin approves or rejects a member's registration             |
| Loan               | Represents loan applications submitted by members                 | Loan application record  | Occurs when member apply for loan                                             |
| applicationDetails | Contains detailed information about the member's loan application | Loan application details | Occurs when a member submits all required information for a loan application  |
| loanApproval       | Represents the approval process for loans submitted by members    | Loan approval process    | Occurs when the admin approves or rejects a loan application                  |
| alert              | Represents automated alerts sent to admins                        | System alerts, reminder  | Occurs when pending approvals or deadlines require action within a set period |

### 2.3.2 Identifying Relationship Types

| Entity Name        | Multiplicity | Relationship         | Entity Name                                | Multiplicity |
|--------------------|--------------|----------------------|--------------------------------------------|--------------|
| Employee           | 1..1         | contain              | memberApplicationDetails                   | 1..*         |
| Member             | 1..1         | view                 | applicationDetails                         | 1..*         |
| Admin              | 1..1<br>1..1 | Validate<br>validate | memberApproval<br>loanApproval             | 1..*<br>1..* |
| memberRegistration | 1..1<br>1..1 | View<br>has          | memberApplicationDetails<br>memberApproval | 1..*<br>1..* |
| Loan               | 1..1<br>1..1 | Contain<br>has       | applicationDetails<br>loanApproval         | 1..*<br>1..* |



|                          |      |      |       |      |
|--------------------------|------|------|-------|------|
| memberApplicationDetails | 1..1 | give | alert | 1..* |
| applicationDetails       | 1..1 | give | alert | 1..* |

### 2.3.3 Attributes Description

| Entity Name | Attributes      | Data Type & Length | Description                         | Nulls | Multivalued | Example          |
|-------------|-----------------|--------------------|-------------------------------------|-------|-------------|------------------|
| Employee    | employeeID (PK) | INT (4)            | Unique ID for the user              | No    | No          | 1234             |
|             | password        | VARCHAR (30)       | User's account password             | No    | No          | P12345W@@        |
| Member      | employeeID (PK) | INT (4)            | Unique ID for member                | No    | No          | M0000001         |
|             | noPF            | INT (4)            | Unique PF number for member         | No    | No          | 1234             |
|             | memberName      | VARCHAR (80)       | Name of the member                  | No    | No          | Ali Abu          |
|             | id              | CHAR (12)          | Identification number of the member | No    | No          | 040101081234     |
|             | email           | VARCHAR (30)       | Member's email address              | No    | No          | member@gmail.com |
|             | phoneNumber     | VARCHAR (15)       | Contact number of the member        | No    | No          | +60123456789     |
|             | sex             | CHAR (1)           | Gender of the member (M/F)          | No    | No          | F                |
|             | religion        | VARCHAR (30)       | Religion of the member              | Yes   | No          | Buddha           |
|             | maritalStatus   | VARCHAR (15)       | Marital status                      | Yes   | No          | Single           |

|                           |                           |                |                                       |     |    |               |
|---------------------------|---------------------------|----------------|---------------------------------------|-----|----|---------------|
|                           | nation                    | VARCHAR (30)   | Nationality of the member             | Yes | No | Malaysia      |
|                           | position                  | VARCHAR (30)   | Working position of the member        | No  | No | Data Engineer |
|                           | monthlySalary             | DECIMAL (10,2) | Member's monthly salary               | No  | No | 10000.00      |
| Admin                     | employeeID (PK)           | INT (4)        | Unique ID for admin                   | No  | No | 1234          |
|                           | adminPassword             | VARCHAR (30)   | User's account password               | No  | No | P12345W@@     |
| memberRegistration        | memberRegistrationID (PK) | VARCHAR (20)   | Unique ID for member registration     | No  | No | MR123456789   |
| memberApplication Details | regisDate                 | DATE           | Date of member application submission | No  | No | 15-01-2025    |
|                           | regisStatus               | VARCHAR (20)   | Member registration status            | No  | No | Pending       |
|                           | regisApproveDeadline      | DATE           | Deadline for admin approval           | No  | No | 25-01-2025    |
| memberApproval            | regisApproveDate          | DATE           | Date of member registration approval  | No  | No | 15-01-2025    |
|                           | regisStatus               | VARCHAR (20)   | Member registration status            | No  | No | Approved      |
| loan                      | loanApplicationID (PK)    | VARCHAR (20)   | Unique ID for loan application        | No  | No | LA123456789   |
|                           | financialType             | VARCHAR (30)   | Type of financing                     | No  | No | Personal Loan |

|                    |                     |                |                                     |    |    |                            |
|--------------------|---------------------|----------------|-------------------------------------|----|----|----------------------------|
|                    | amountRequested     | DECIMAL (12,2) | Amount requested for the loan       | No | No | 30000.00                   |
|                    | financingPeriod     | INT            | Loan repayment period in months     | No | No | 24                         |
|                    | monthlyInstallments | DECIMAL (10,2) | Monthly repayment amount            | No | No | 650.00                     |
|                    | bankName            | TEXT           | Name of the bank                    | No | No | Maybank                    |
|                    | bankAcc             | INT            | Bank account number                 | No | No | 123456789                  |
| applicationDetails | applicationDate     | DATE           | Date of loan application submission | No | No | 15-01-2025                 |
|                    | loanStatus          | VARCHAR (20)   | Status of the loan application      | No | No | Pending                    |
|                    | loanApproveDeadline | DATE           | Deadline for admin approval         | No | No | 25-01-2025                 |
| loanApproval       | loanApproveDate     | DATE           | Date of loan application approval   | No | No | 15-01-2025                 |
|                    | loanStatus          | VARCHAR (20)   | Status of loan application          | No | No | Approved                   |
| alert              | alertID             | VARCHAR (20)   | Unique ID for alerts                | No | No | 200001                     |
|                    | alertType           | VARCHAR (20)   | Type of alert                       | No | No | Deadline reminder          |
|                    | alertContent        | TEXT           | Content of the alert                | No | No | Your registration validate |

|  |  |  |  |  |  |                          |
|--|--|--|--|--|--|--------------------------|
|  |  |  |  |  |  | deadline is approaching. |
|--|--|--|--|--|--|--------------------------|

## 2.4 Relational Database Schema

EMPLOYEE (employeeID, password)

PK: employeeID

ADMIN (employeeID, adminPassword)

PK: employeeID

FK: employeeID references EMPLOYEE (employeeID)

MEMBER (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary)

PK: employeeID

FK: employeeID references EMPLOYEE (employeeID)

LOAN(loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAccount)

PK: loanApplicationID

APPLICATIONDETAILS (employeeID, loanApplicationID, applicationDate, loanStatus, loanApproveDeadline)

PK: employeeID, loanApplicationID

FK: employeeID references EMPLOYEE (employeeID)

FK: loanApplicationID references LOAN (loanApplicationID)

LOANAPPROVAL (employeeID, loanApplicationID, loanApproveDate, loanStatus)

PK: employeeID, loanApplicationID

FK: employeeID references EMPLOYEE (employeeID)

FK: loanApplicationID references LOAN (loanApplicationID)

MEMBERREGISTRATION (memberRegistrationID)

PK: memberRegistrationID

MEMBERAPPLICATIONDETAILS (employeeID, memberRegistrationID, regisDate, regisStatus, regisApproveDeadline)

PK: employeeID, memberRegistrationID

FK: employeeID references EMPLOYEE (employeeID)

FK: memberRegistrationID references MEMBERREGISTRATION (memberRegistrationID)

MEMBERAPPROVAL (employeeID, memberRegistrationID, regisApproveDate, regisStatus)

PK: employeeID, memberRegistrationID

FK: employeeID references EMPLOYEE (employeeID)

FK: memberRegistrationID references MEMBERREGISTRATION  
(memberRegistrationID)

## 2.5 Normalization

### **Employee – memberApplicationDetails**

1NF:

Employee\_memberApplicationDetails (employeeID, password, regisDate, regisStatus, regisApproveDeadline)

PK: employeeID

2NF:

Employee (employeeID, password)

PK: employeeID

Employee\_memberApplicationDetails (employeeID, regisDate, regisStatus, regisApproveDeadline)

PK: employeeID references employee(employeeID)

3NF:

Employee (employeeID, password)

PK: employeeID

Employee\_memberApplicationDetails (employeeID, regisDate, regisStatus, regisApproveDeadline)

PK: employeeID references employee(employeeID)

### **memberRegistration – memberApplicationDetails**

1NF:

memberRegistration\_memberApplicationDetails (memberRegistrationID, regisDate, regisStatus, regisApproveDeadline)

PK: memberRegistrationID

2NF:

memberRegistration\_memberApplicationDetails (memberRegistrationID, regisDate, regisStatus, regisApproveDeadline)

PK: memberRegistrationID

3NF:

memberRegistration\_memberApplicationDetails (memberRegistrationID, regisDate, regisStatus, regisApproveDeadline)

PK: memberRegistrationID

### **memberRegistrationID – memberApproval**

1NF:

memberRegistrationID\_memberApproval (memberRegistrationID, regisApproveDate, regisStatus)

PK: memberRegistrationID

2NF:

memberRegistrationID\_memberApproval (memberRegistrationID, regisApproveDate, regisStatus)

PK: memberRegistrationID

3NF:

memberRegistrationID\_memberApproval (memberRegistrationID, regisApproveDate, regisStatus)

PK: memberRegistrationID

### **memberApplicationDetails – Alert**

1NF:

Alert\_memberApproval (alertID, alertType, alertContent, regisDate, regisStatus, regisApproveDeadline)

PK: alertID

2NF:

Alert (alertID, alertType, alertContent)

PK: alertID

Alert\_memberApproval (alertID, regisDate, regisStatus, regisApproveDeadline)

PK: alertID

FK: alertID references alert (alertID)

3NF:

Alert (alertID, alertType, alertContent)

PK: alertID

Alert\_memberApproval (alertID, regisDate, regisStatus, regisApproveDeadline)

PK: alertID

FK: alertID references alert (alertID)

### **Admin – memberApproval**

1NF:

Admin\_memberApproval (employeeID, adminPassword, regisApproveDate, regisStatus)

PK: employeeID

2NF:

Admin (employeeID, adminPassword)

PK: employeeID

Admin\_memberApproval (employeeID, regisApproveDate, regisStatus)

PK: employeeID

FK: employeeID references admin (employeeID)

3NF:

Admin (employeeID, adminPassword)

PK: employeeID

Admin\_memberApproval (employeeID, regisApproveDate, regisStatus)

PK: employeeID

FK: employeeID references admin (employeeID)

### **Admin – loanApproval**

1NF:

Admin\_loanApproval (employeeID, adminPassword, loanApproveDate, loanStatus)

PK: employeeID

2NF:

Admin (employeeID, adminPassword)

PK: employeeID

Admin\_loanApproval (employeeID, loanApproveDate, loanStatus)

PK: employeeID

FK: employeeID references admin (employeeID)

3NF:

Admin (employeeID, adminPassword)

PK: employeeID

Admin\_loanApproval (employeeID, loanApproveDate, loanStatus)

PK: employeeID

FK: employeeID references admin (employeeID)

### **loan – loanApproval**

1NF:

Loan\_loanApproval (loanApplicationID, financialType, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAcc, loanApproveDate, loanStatus)

PK: loanApplicationID

2NF:

Loan (loanApplicationID, financialType, amountRequested, financingPeriod, monthlyInstallments)

PK: loanApplicationID

Loan\_loanApproval (loanApplicationID, bankName, bankAcc, loanApproveDate, loanStatus)

PK: loanApplicationID

FK: loanApplicationID references loan (loanApplicationID)

3NF:

Loan (loanApplicationID, financialType, amountRequested, financingPeriod, monthlyInstallments)

PK: loanApplicationID

Bank (bankName, bankAcc)

PK: bankAcc

Loan\_loanApproval (loanApplicationID, bankAcc, loanApproveDate, loanStatus)

PK: loanApplicationID, bankAcc

FK: loanApplicationID references loan (loanApplicationID)

FK: bankAcc references bank (bankAcc)

### **Loan – applicationDetails**

1NF:

Loan\_applicationDetails (loanApplicationID, financialType, amountRequested, financingPeriod, monthlyInstallments, applicationDate, loanStatus, loanApproveDeadline)

PK: loanApplicationID

2NF:

Loan (loanApplicationID, financialType, amountRequested, financingPeriod, monthlyInstallments)

PK: loanApplicationID

Loan\_applicationDetails (loanApplicationID, applicationDate, loanStatus, loanApproveDeadline)

PK: loanApplicationID

FK: loanApplicationID references loan (loanApplicationID)

3NF:

Loan (loanApplicationID, financialType, amountRequested, financingPeriod, monthlyInstallments)

PK: loanApplicationID

Loan\_applicationDetails (loanApplicationID, applicationDate, loanStatus, loanApproveDeadline)

PK: loanApplicationID

FK: loanApplicationID references loan (loanApplicationID)

### **Member – applicationDetails**

1NF:

Member\_applicationDetails (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, applicationDate, loanStatus, loanApproveDeadline)

PK: employeeID

2NF:

Member (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

Member\_applicationDetails (employeeID, applicationDate, loanStatus, loanApproveDeadline)

PK: employeeID

FK: employeeID references member (employeeID)

3NF:

Member (employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

Member\_applicationDetails (employeeID, applicationDate, loanStatus,



loanApproveDeadline)  
PK: employeeID  
FK: employeeID references member (employeeID)

**applicationDetails – alert**

1NF:  
alert\_applicationDetails (alertID, alertType, alertContent, applicationDate, loanStatus, loanApproveDeadline)  
PK: alertID

2NF:  
Alert (alertID, alertType, alertContent)  
PK: alertID  
alert\_applicationDetails (alertID, applicationDate, loanStatus, loanApproveDeadline)  
PK: alertID  
FK: alertID references alert (alertID)

3NF:  
Alert (alertID, alertType, alertContent)  
PK: alertID  
alert\_applicationDetails (alertID, applicationDate, loanStatus, loanApproveDeadline)  
PK: alertID  
FK: alertID references alert (alertID)